

Ravani Roshan

AI/ML Engineer · LLM Fine-Tuning Developer · Applied ML Systems Builder
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PROFESSIONAL SUMMARY

Computer Science Engineering undergraduate focused on applied machine learning, LLM fine-tuning, transformer systems, RAG pipelines, and production-oriented ML infrastructure. Built projects including a physics-informed solar flare early-warning system, a pure NumPy GPT-2/RoPE implementation, a cognitive telemetry framework for memory-augmented agents, and a hermetic multi-agent coding system. Strong hands-on with Python, Rust, NumPy, FastAPI, Hugging Face, LangChain, QLoRA/DPO, Docker, and RAG.

EDUCATION

Silver Oak University

B.Tech, Computer Science Engineering

Ahmedabad, Gujarat, India

CGPA: 8.0

EXPERIENCE

Google Developers Group

Core Developer

Oct 2023 – Dec 2025

Ahmedabad, Gujarat, India

- Led and supported technical sessions for a 200+ developer community on AI/ML tooling, LLM integration, RAG systems, and deployment practices.
- Mentored peers building applied AI systems using Python, ML APIs, LangChain/LangGraph, and production deployment workflows.
- Contributed to community projects across applied ML, AI tooling, and automation systems.

PROJECTS

Niki – Hermetic Multi-Agent Coding System | *Rust, Docker, CLI, LLM, Multi-agent*

Jul 2026 – Present

- Built an autonomous multi-agent system for code generation: agents reason through problem decomposition, implement solutions, run tests, and conduct peer review—all in isolated Docker sandboxes.
- Designed agent orchestration with LLM-agnostic routing, structured reasoning chains, and safety-first sandbox isolation.

ArkaNetra – Physics-Informed Solar Flare Early Warning | *Python, ML, Docker*

Jun 2026 – Present

- Built a physics-informed multimodal solar flare early-warning demo with flare probability, model confidence, explainability, anomaly detection, and downstream risk assessment.
- Structured as a production-style Python ML project with modules, configs, data, models, tests, Docker Compose, and reproducible commands.

Transformer from Scratch – Pure NumPy GPT-2/RoPE | *Python, NumPy, Transformers*

May 2026 – Present

- Implemented a GPT-2-style decoder-only Transformer in pure NumPy: multi-head attention, RoPE embeddings, GELU activations, residual blocks, and autoregressive sampling.

Mnemo – Cognitive Telemetry for LLM Agents | *Python, LLM Agents, Observability*

May 2026 – Present

- Built telemetry infrastructure for memory-augmented agents: trace recording, failure attribution, belief tracking, memory replay, and metrics reporting.

TECHNICAL SKILLS

Machine Learning: Supervised ML, model evaluation, uncertainty estimation, anomaly detection, explainability, CV, NLP, transformers

LLM/GenAI: LLM fine-tuning, QLoRA, DPO, synthetic data, RAG, LangChain, LangGraph, prompt engineering, LLM observability

Programming: Python, Rust, NumPy, JavaScript/TypeScript, C++, C# fundamentals

ML Engineering: FastAPI model serving, Hugging Face Transformers, Docker, reproducible pipelines, REST APIs, WebSockets

Data/Cloud: PostgreSQL, Supabase, SQLite, MySQL, Vercel, Railway, Fly.io, AWS, GCP, GitHub Actions

CERTIFICATIONS

Google: Machine Learning Operations (MLOps) for GenAI (Dec 2025), Transformer Models and BERT Model (Mar 2026), Responsible AI: Privacy & Safety (Jan 2026)

DeepLearning.AI: Finetuning Large Language Models, Prompt Engineering for Vision Models (Dec 2024), LangChain for LLM Application Development

Other: Machine Learning with Python (freeCodeCamp / Cognitive Class, Dec 2024), Computer Vision (Kaggle, Nov 2024)